

ATHENEA

Relightable Gaussians from production 3D

A relight baker: USD models become Gaussian splats that can be relit under new light and still match a path-traced reference.

jesus luque · 12 October 2026

Faithful images are slow; fast ones are frozen



Left: the mesh, path traced. Right: the same pawn as Gaussians.

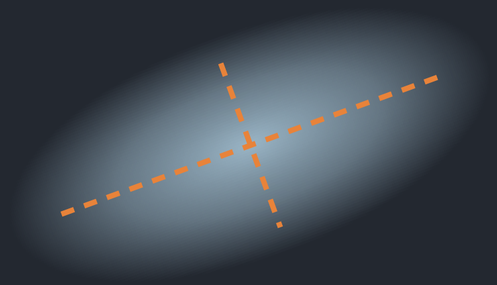
Path tracing

Follows light physically, so it is the faithful reference. Too slow to use interactively.

Gaussian splats

Fast and photographic, but usually trained from photos: the light is fixed, with no materials and no link to the 3D asset.

What a Gaussian splat is



One splat

A soft, coloured, see-through ellipse in 3D: a position, a size and shape, a direction, an opacity and a colour.



Millions of them

Sorted and blended on screen, they add up to a photographic image, fast enough to move around in.

Bake light once, relight anywhere

1 · BAKE

Trace the light

A GPU path tracer works out, for every Gaussian, how light reaches it and leaves it: which directions are open, what it reflects.

2 · STORE

Keep it in each splat

That light transport is saved inside the Gaussian, next to its colour and material.

3 · RELIGHT

Any sky, any light

When the model is drawn, a new sky or light is applied in a quick step. No path tracing at that point.

The expensive work happens once, offline. Drawing stays fast.

The pipeline in five steps

1

USD asset

Meshes, materials, textures and, for characters, a skeleton.

→

2

Gaussians

The surface becomes splats, each about one pixel of a chosen camera.

→

3

Bake

The path tracer stores in each splat how light moves on it.

→

4

Levels of detail

Splats merge into trees, so a far view draws fewer.

→

5

Player

The cloud is drawn and relit under the sky of choice.

Every step that touches scene data runs on the GPU.

THE YARDSTICK

Always compared with a path trace of the original model

Same camera, same sky. Image against image, region by region. Where quality and speed conflict, quality comes first.

Finer bake, closer to the path trace



Path trace
the reference

Bake at 640 px
soft, smeared detail

Bake at 1280 px
sharper detail

CAR CROP SCORE

17.88 dB

up from 16.55 dB

The score (PSNR) measures how close the image is to the path trace; higher is closer. Doubling the bake's resolution raised it by 1.3 dB.

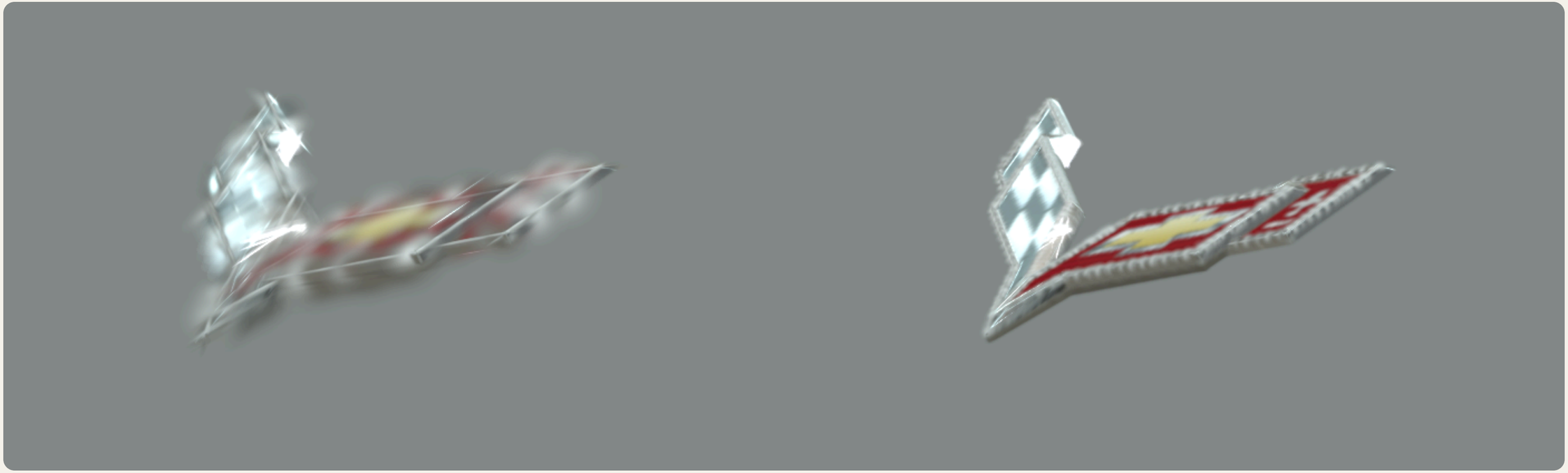


WHOLE FRAME

23.90 dB

Up from 22.18 dB at 640 px. The car is drawn as relit Gaussians, scored against the path trace from the same camera and sky.

Close-ups get their own finer level



Baked at 1280 px

Fine from a normal distance, soft up close.

Baked at 5120 px

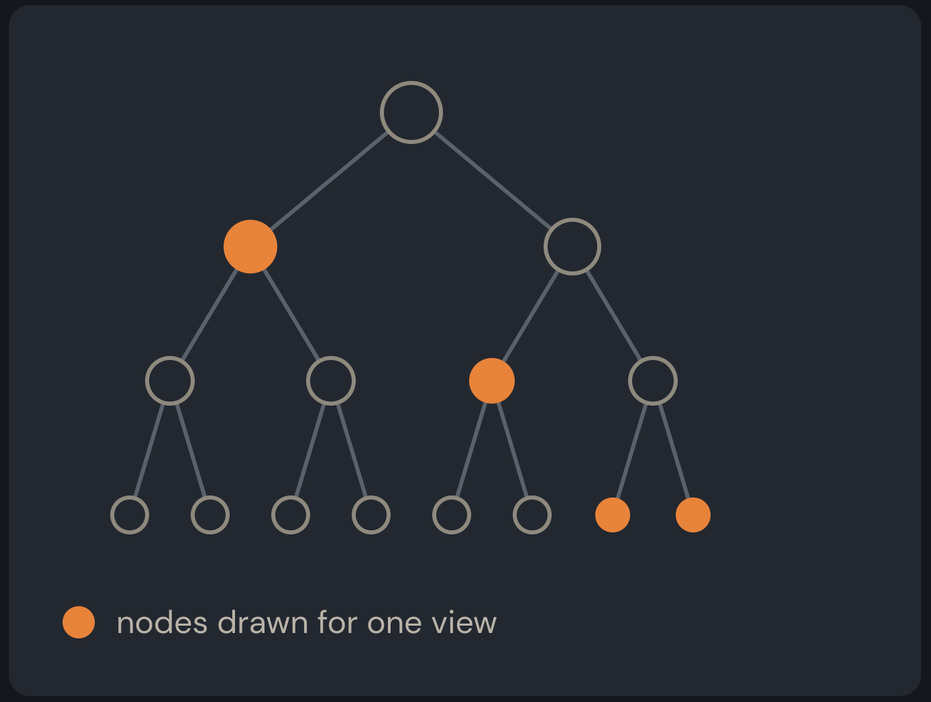
The checkered flag and the bowtie come out crisp.

Levels of detail: draw what the view needs

Neighbouring Gaussians are merged pair by pair into a tree. For each view, the player picks the coarsest nodes whose error on screen stays under a pixel, and skips backs and hidden parts.

1/8
of the Gaussians drawn

~0.1 dB
lost against the path trace



Animated assets keep their skeleton



The rig travels

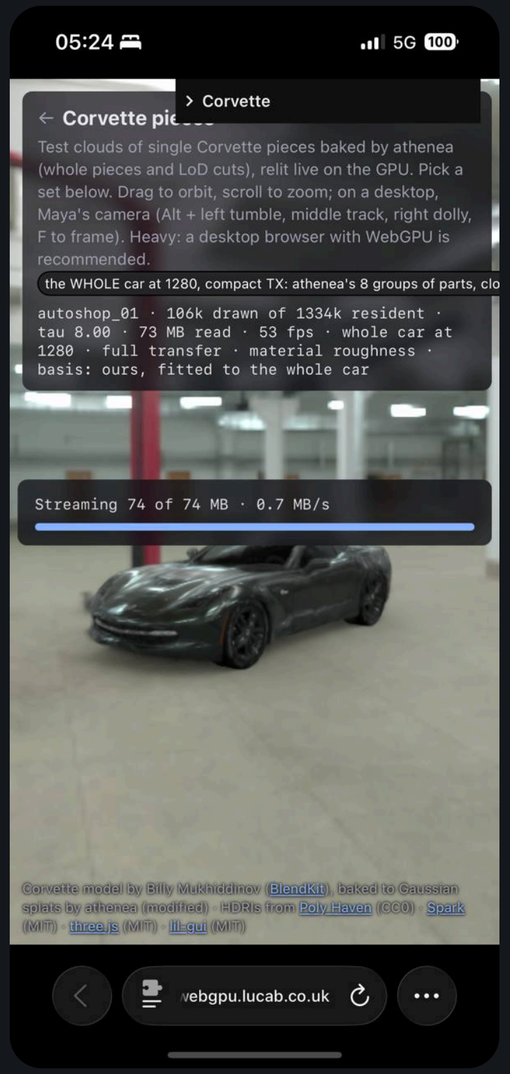
Splats are stored once, with the bones that move them, instead of once per frame.

Light turns with a limb

The baked light rotates with each part as the sparrow flies.

The pawn as control

A static object with no skeleton: every change for animation must leave it unchanged.



A PARALLEL APPLICATION

The same car in a phone browser

Smooth on a real iPhone

The whole relit car, compressed, drew 106 thousand Gaussians at 53 frames per second in the default view.

Sharper when you stop

When the camera rests, the player draws one finer frame, then waits for the next touch.

Close-ups were soft

The cause was the 1280 px bake, not the player. That led to the 5120 px close-up level.

What is next

- **Lamps:** headlights come out dark; their light is not baked yet.
- **Thin chrome:** the bumper's strip is 38 % brighter than the path trace.
- **Hood highlights** sit 2 px off their place.
- **One car, one sky** measured so far; other assets and light to follow.
- **On the phone,** the 5120 px close-up level is still to be measured.



Chrome strip: path trace (top), two bakes below.

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From production USD to Gaussians that relight

Convert → Bake light → Relight → Check against the path trace

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